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Article

The Reinforcement Learning Contour: From Document Production to Governed Epistemic Transformation in AI-Assisted Scientific Inquiry

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Abstract

Generative artificial intelligence has sharply reduced the technical cost of producing scientific text. Increased production capacity does not, however, entail cumulative knowledge, conceptual novelty, or epistemic progress. This paper introduces the Reinforcement Learning Contour (RLC) as a conceptual and architectural framework for shifting AI-assisted research from repetitive content generation toward governed recursive learning. The RLC is not proposed as a conventional reinforcement-learning algorithm and does not claim equivalence with a Markov decision process, scalar reward function, value function, or policy-optimization procedure. The framework distinguishes two analytical levels: an *epistemic contour* is the time-indexed state of a bounded configuration of knowledge, ontology, relational structure, retrieval mechanisms, evaluative criteria, inquiry policy, provenance, memory, tools, and governance constraints; the *Reinforcement Learning Contour* is the governed recursive architecture that links such states through evidence, action, evaluation, approval, and reintegration. The central proposition is a differentiation principle: the value of a research cycle is proportional not to the volume of output it generates, but to the beneficial and traceable differentiation it produces between the preceding epistemic contour and the contours governing subsequent perception, interpretation, decision, and action. This reframes the unit of scientific analysis from the document to the transformation of the mechanism that produced the document. The paper positions the principle relative to Lakatos's progressive and degenerating problem shifts, Popperian criticism, and double-loop learning; specifies a role-based architecture in which provenance, temporality, and human acceptance authority are structural rather than procedural; provides a Structured Epistemic Change Record that documents differentiation without reducing scientific quality to a single optimized score; enumerates failure modes and controls; and states seven falsifiable hypotheses with a comparative evaluation design. One local implementation is described as an existence proof, combining a prior-research corpus, a governed knowledge store, a temporal provenance graph, source-grounded synthesis, adversarial external discovery, and local orchestration under sole human acceptance authority. The framework is explicitly presented as an untested conceptual proposal. A research cycle may be epistemically successful even when it produces no publishable document, provided it improves the mechanism from which future inquiry emerges.

Keywords: reinforcement learning contour; AI-assisted scientific inquiry; epistemic transformation; research programmes; double-loop learning; provenance; temporal knowledge graphs; agentic science; human governance; scientific methodology

1. Introduction

Large language models can summarize literature, generate hypotheses, propose research designs, draft manuscripts, and reorganize complex arguments. Autonomous and semi-autonomous research agents extend these capabilities toward experimental planning and end-to-end scientific artifact production. Yet productivity and progress are not equivalent. A system may produce a new

manuscript every day while preserving the same assumptions, conceptual vocabulary, blind spots, and citation errors. Fluency can create an illusion of understanding, in which apparent breadth substitutes for depth of comprehension and the space of questions actually asked silently narrows [22]. Recursive dependence on generated outputs can additionally narrow the represented distribution and reinforce a self-confirming programme, a degradation demonstrated directly for models trained on recursively generated data [21].

The core problem is therefore not how to automate document production. It is how to ensure that each research episode modifies, in a beneficial and inspectable way, the epistemic mechanism from which subsequent inquiry begins.

This paper proposes the Reinforcement Learning Contour (RLC) as a conceptual and architectural response. The RLC is not used as a synonym for a conventional reinforcement-learning algorithm. No claim is made that the proposed system constitutes a Markov decision process, learns an optimal policy from a scalar reward, or implements a value function in the technical sense. The construct instead extends the logic of recursive feedback and policy transformation to governed scientific inquiry; the relationship to technical reinforcement learning is stated precisely in Section 3.1.

The framework is motivated by four failure conditions. Textual novelty can occur without epistemic novelty; a closed corpus can become increasingly coherent and decreasingly corrigible; unmanaged external expansion can replace internal bias with unverified novelty; and a document can be treated as an endpoint rather than as an epistemic event that should modify future inquiry.

The paper makes five claims to contribution. First, it relocates the unit of analysis from the artifact to the generative mechanism and operationalizes that relocation. Second, it separates the epistemic contour as a state representation from the RLC as the recursive governance architecture that transforms one state into another. Third, it specifies a role-level architecture in which provenance, temporality, and human acceptance authority are structural rather than procedural. Fourth, it provides a Structured Epistemic Change Record together with the conditions under which that record is invalid. Fifth, it states falsifiable hypotheses and a comparative design so that the framework can be refuted rather than merely adopted.

2. Philosophical and Technical Foundations

2.1. Popper: Criticism and Exposure to Refutation

Popper's account of scientific knowledge emphasizes conjecture, criticism, and exposure to possible refutation [1,2]. Within the RLC architecture, this principle is operationalized not as a universal numerical score but as an adversarial stage: the system must actively retrieve disconfirming evidence, formulate conditions under which a central claim would fail, and distinguish an explanatory proposal from an empirically validated conclusion.

2.2. Lakatos: Progressive and Degenerating Research Programmes

Lakatos provides the closest philosophical predecessor to beneficial contour differentiation. A research programme is progressive when theoretical change generates novel empirical content and is degenerating when modifications merely protect an established core after difficulties arise [3]. The RLC does not claim to rediscover this distinction. Its proposed contribution is operational: representing a research programme as a sequence of inspectable epistemic state transitions with claim-level lineage, temporal provenance, contradiction records, ontology changes, and explicit human approval.

The relationship can be stated as follows: Lakatos asks when a research programme is progressing; the RLC asks how a human-machine research environment can record, govern, and compare the transitions through which such progress or degeneration may occur. Lakatos's appraisal was retrospective and historiographic, offering no mechanism for making a shift visible at the time it

occurs. The architecture proposed here attempts to make differentiation contemporaneous and inspectable.

2.3. Kuhn, Laudan, and Socially Situated Inquiry

Kuhn's paradigms, Laudan's problem-solving model, and Longino's account of socially mediated objectivity further caution against reducing scientific change to a single metric [4–6]. Scientific progress may involve conceptual reclassification, improved problem-solving, exposure of background assumptions, or stronger critical interaction. Consequently, the RLC treats epistemic transformation as multidimensional and contestable rather than as a scalar reward automatically produced by the system. Longino's argument is additionally relevant to Section 10: objectivity is a property of critical interaction among differently situated inquirers, which a single-researcher recursive system structurally lacks.

2.4. Single-Loop and Double-Loop Learning

Argyris and Schön distinguished single-loop learning, which corrects action within existing governing variables, from double-loop learning, which revises the governing variables themselves [7]. The RLC's distinction between weak learning (changing an answer) and strong learning (changing the function that produces answers) is a restatement of that distinction in an epistemic-systems register, and is acknowledged as such rather than presented as new. The proposed contribution is not the distinction itself but the attempt to instrument it: to specify the snapshots, provenance records, difference reports, and authority separations required for a claim of strong learning to be auditable rather than merely asserted.

2.5. Provenance, Reproducibility, and Machine-Supported Inquiry

Modern scientific workflows depend on traceable data, methods, software, claims, and transformations. The W3C PROV model formalizes the entities, activities, and agents involved in producing information [8]. FAIR principles emphasize findability, accessibility, interoperability, and reuse [9]. Reproducible research similarly requires that computational results remain connected to executable procedures and inspectable evidence [10,11], and documentation practice has extended this from data through software to provenance itself [12]. Foundational work on data provenance established the distinction between *why* and *where* lineage and surveyed its representation in e-science [13,14].

The RLC extends these concerns along one axis: from reproducibility of an output to provenance of the epistemic changes that shape future inquiry. A workflow may be fully reproducible and still leave no record of why the research programme now asks different questions than it did a month ago.

2.6. Knowledge Representation, Grounding, and Citation Integrity

Static knowledge graphs represent that a source supports a claim [15]; they do not natively represent that the claim was later narrowed, contradicted, or superseded. Dynamic and temporal graph research addresses this through validity intervals and relation evolution [16,17]. Retrieval-augmented generation conditions output on retrieved documents [18], which mitigates but does not eliminate unsupported assertion; hallucination remains a documented general property of natural language generation [19]. Most directly relevant here, language models fabricate plausible bibliographic references at non-trivial rates [20].

This last finding is load-bearing rather than incidental. A recursive system that ingests its own unverified outputs converts individual fabrications into structural corruption of the knowledge base, and the corruption compounds silently across cycles because the surrounding prose remains fluent.

2.7. Agentic Science and the Gap Addressed Here

Autonomous and semi-autonomous research systems establish the feasibility of increasingly complete research cycles, from experimental planning and execution to end-to-end manuscript generation [24–26]. Scholarship on large models in science warns in parallel that fluency, scale, and apparent breadth can be mistaken for understanding, and that such capability requires explicit epistemic governance rather than tool-level guidelines [22,27,28].

Neither line addresses the question posed here, which is not whether such systems can produce scientific artifacts but whether the resulting sequence of artifacts constitutes a progressive, traceable, and corrigible programme. The RLC is positioned as a complementary appraisal and governance layer, applicable to such systems including as a criterion for evaluating them.

3. Definition of the Reinforcement Learning Contour

The framework operates on two analytical levels, and keeping them apart is what makes the construct precise.

An **epistemic contour** is the time-indexed state C_t of a bounded configuration of knowledge, ontology, relational structure, retrieval mechanisms, evaluative criteria, inquiry policies, memory, provenance, available tools, and governance constraints. It determines how a human-machine system perceives evidence, constructs questions, selects actions, and evaluates outcomes at a given moment.

$$C_t = \langle K_t, O_t, G_t, R_t, P_t, E_t, M_t, H_t \rangle$$

Here K denotes available knowledge; O the active ontology; G the graph of entities, claims, and relations; R retrieval mechanisms; P inquiry and action policies; E evaluative criteria; M memory and provenance; and H human and institutional governance.

The **Reinforcement Learning Contour** is the governed recursive architecture that links such states to one another through evidence, action, evaluation, approval, and reintegration. The epistemic contour is what the system is at time t ; the RLC is what carries it from t to $t+1$ under governance. This distinction preserves the descriptive precision of the state representation while retaining RLC as the name of the complete learning system, and it locates the word *reinforcement* where it belongs: in the mechanism of transition, not in the static configuration.

A research episode X_t produces not only a document D_t but a possible transition from C_t to C_{t+1} . The document is one artifact of the episode. The primary object of evaluation is the change in the mechanism governing future inquiry.

$$C_t \rightarrow [X_t] C_{t+1}$$

3.1. Relation to Technical Reinforcement Learning

The name Reinforcement Learning Contour is retained deliberately, but the relationship to technical reinforcement learning is structural and analogical rather than formally algorithmic. Conventional reinforcement learning presupposes a defined state space, action space, reward signal, and policy updated to maximize expected return [29]. The RLC has an open and partly unformalized state space; its evaluative criteria are multidimensional, contested, and often delayed; and its updates include human judgment that is not reducible to optimization of a scalar objective. The retained term marks the recursive principle that evaluated outcomes modify the conditions and policies governing future action. It does not claim that the architecture itself is an RL algorithm.

The word *contour* is used in an architectural sense: a bounded, recursively connected feedback configuration whose internal relations and boundary conditions evolve over time, through which prior states constrain and are modified by later states. This usage is distinct from the geometric meaning of an outline or isoline. The term is retained because the object of interest is not a single loop operation, but the evolving boundary and internal organization of the entire knowledge-action configuration. The terminology therefore names a broader epistemic system inspired by recursive reinforcement, while explicitly refusing technical equivalence with a Markov decision process.

Where reinforcement-learning concepts do apply directly, they apply as warnings rather than as machinery. Reward hacking [23] is a live hazard for any scored novelty criterion, and Section 7 is constructed specifically to avoid creating that surface.

4. The Law of Experiential Value

The central proposition of the framework is:

The greater the beneficial differentiation between a preceding epistemic contour and the contours governing future perception, decision, and action, the greater the value of the lived or computationally mediated experience.

The differentiation is between states; the Reinforcement Learning Contour is the architecture within which that differentiation is produced, recorded, and judged. The qualifier *beneficial* is indispensable. Any episode may alter a system, but alteration may produce confusion, false confidence, epistemic closure, degraded calibration, or stronger attachment to unsupported claims. Change alone is not learning. D+ is accordingly a relational judgment supported by evidence and governance records, not a scalar reward to be optimized.

$$V(X_t) \propto D^+(C_t, C_{t+1})$$

D+ should not be interpreted as an already validated numerical metric. It denotes beneficial differentiation established through explicit evidence and judgment, recorded in the form specified in Section 7. The value of an episode is high when it changes not merely an answer, but the mechanism that generates future answers.

Weak learning: $A_t \rightarrow A_{t+1}$

Strong learning: $f_t(\cdot) \rightarrow f_{t+1}(\cdot)$

The architectural claim attached to this distinction is narrow and testable: an auditable claim of strong learning requires, at minimum, that contour state be snapshotted, that differences be recorded with provenance, and that acceptance be separated from generation.

5. Architecture by Functional Roles

To avoid dependence on short-lived products, the architecture is defined by roles rather than brands. The authority column is the load-bearing element: it specifies what each role may commit, not merely what it may do.

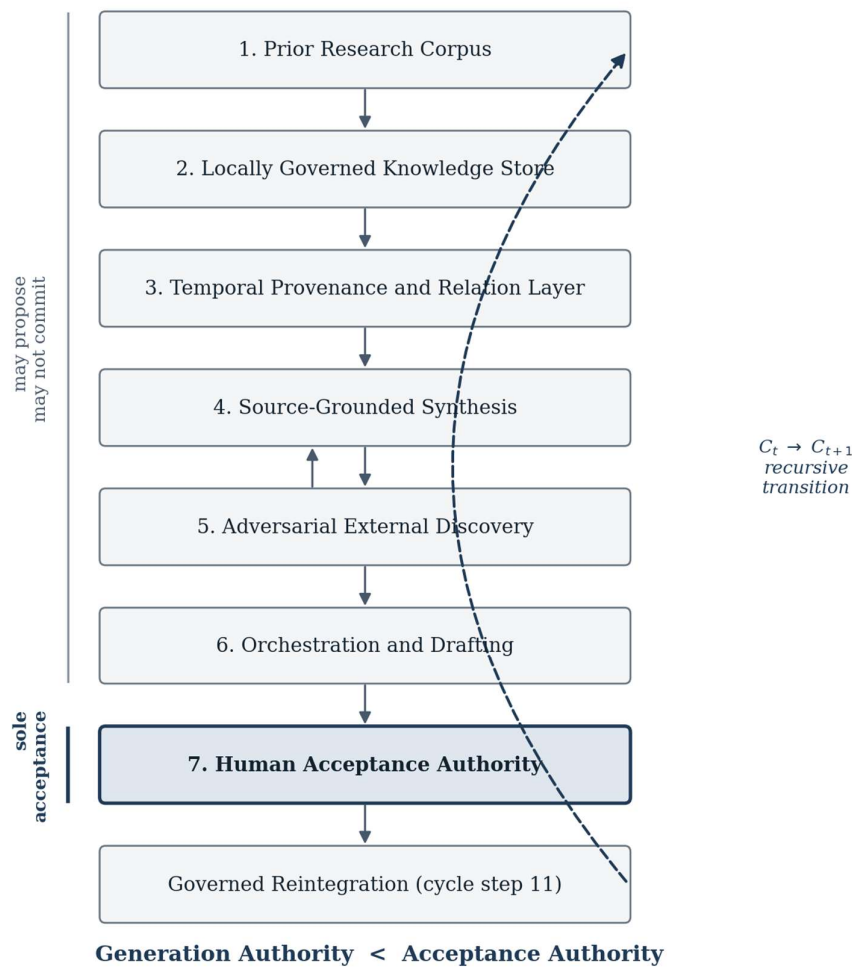


Figure 1. *The Reinforcement Learning Contour as a governed recursive architecture. Roles 1–6 may propose changes to the epistemic contour; role 7 alone may commit them. The dashed return path marks the transition from C_t to C_{t+1} , which is the object of evaluation rather than the document produced along the way.*

Architectural role	Function	Authority
Prior Research Corpus	Publications, manuscripts, notes, reviewer feedback, rejected hypotheses, methods, and unresolved questions.	Epistemic memory; no authority
Locally Governed Knowledge Store	Human-readable and editable representation of concepts, claims, evidence, decisions, and research questions.	Working representation
Temporal Provenance and Relation Layer	Versioned entities and relations, validity intervals, source lineage, contradiction and supersession records.	System of record for change

Source-Grounded Synthesis Layer	Analysis restricted to approved evidence, cross-document comparison, claim extraction, and structured synthesis.	May propose; may not commit
External Discovery and Adversarial Retrieval Layer	Search for new literature, competing theories, negative evidence, replications, criticism, and adjacent domains.	May propose; may not commit
Research Orchestration and Drafting Layer	Workflow execution, retrieval coordination, structured drafting, validation scripts, and change reports.	May propose; may not commit
Human Epistemic Governance	Approval of questions, evidence, ontology changes, interpretations, authorship, and publication.	Sole acceptance authority; veto

Generation authority is strictly weaker than acceptance authority, and acceptance is held exclusively by the human governance role. The synthesis, discovery, and orchestration layers may propose RLC changes; none may commit them. A configuration in which the drafting layer writes to the temporal provenance layer without an approval event has abandoned the architecture regardless of the sophistication of its tooling. This allocation is what makes the separation demanded in Section 10 structural rather than merely procedural.

Reintegration of validated knowledge is a step of the research cycle (Section 6, step 11) rather than an architectural role. It is executed by the orchestration layer under an approval event issued by the human governance role.

5.1. Requirements per Role

Prior Research Corpus. Each item stored with version, status, domain, extracted claims, evidence type, confidence, citations, known contradictions, and relation to subsequent work. Rejected and falsified material must be retained; a corpus that stores only successes cannot support calibration.

Locally Governed Knowledge Store. Typed node vocabulary, minimally: Concept, Claim, Hypothesis, Evidence, Method, Dataset, Contradiction, Limitation, Research Question, Publication, External Source, Decision, Contour Update. Links are typed and directional. The store must remain editable by the human without tooling mediation, which is the fallback when automation fails.

Temporal Provenance and Relation Layer. Must represent not only what is believed but when it came to be believed, on what source, what subsequently changed, and whether the earlier claim remains active. Supersession is an explicit relation, not a deletion [16,17]. Without this, differentiation cannot be computed at all, only asserted. Claim records follow the form: Claim = <content, status, confidence, source, time, verification>.

Source-Grounded Synthesis Layer. Value derives from controllable scope, not from authority. Analysis must not silently expand beyond the approved corpus.

External Discovery and Adversarial Retrieval Layer. Mandated to introduce challenge, not merely addition. A discovery pass returning only confirmatory material should be treated as a failed pass and re-run under adversarial framing. The pipeline runs: discovery, original source, verification, claim extraction, provenance record, human decision.

Research Orchestration and Drafting Layer. Must produce a machine-readable change record per cycle. May not write accepted claims to the provenance layer without an approval event.

Human Epistemic Governance. No automated output overrides rejection. The researcher may reject a source, an ontology update, a causal inference, a manuscript, or the direction of the programme.

5.2. One Local Instantiation

The author’s current local implementation uses the following products as one replaceable instantiation of the functional architecture. These products are not part of the definition of the RLC.

Role	Current implementation
Locally governed knowledge store	Obsidian [32]
Temporal provenance and relation layer	Graphiti [33]
Source-grounded synthesis layer	NotebookLM [34]
External discovery and adversarial retrieval	Perplexity [35]
Research orchestration and drafting	Codex [36]
Final epistemic authority	Human researcher

Any substitution preserving the authority allocation of Section 5 is an equivalent instantiation. Any configuration collapsing the synthesis, discovery, and orchestration layers into a single unaudited agent, or permitting automated writes to the temporal provenance layer, is not.

6. The Governed Research Cycle

1. **Contour snapshot.** Record the current knowledge, ontology, graph, policies, criteria, provenance state, and governance constraints. Without this, all subsequent differentiation claims are unverifiable.
2. **Question selection.** Select questions from unresolved contradictions, weak conceptual links, failed hypotheses, external developments, or human priorities.
3. **Internal retrieval.** Retrieve relevant prior publications, claims, evidence, methods, decisions, and limitations.
4. **Source-grounded synthesis.** Analyze the selected internal corpus without silently expanding beyond it.
5. **External and adversarial discovery.** Search for current evidence, competing explanations, criticism, replication failures, and adjacent concepts. Adversarial framing is mandatory.
6. **Primary-source verification.** Verify claims against original sources; search summaries remain discovery objects rather than evidence objects. The verification standard is specified in Section 9.2.
7. **Change proposal.** Specify what would change in the RLC: claim, definition, relation, evidence weight, uncertainty, policy, or governance rule, and on what basis.
8. **Drafting.** Generate a manuscript or other scientific artifact while separating evidence, interpretation, hypothesis, and speculation.
9. **Adversarial evaluation.** Attempt to refute the central claim, detect prior art, identify overgeneralization, and test citation integrity.
10. **Human decision.** Approve, revise, defer, or reject the proposed artifact and RLC changes. Rejection is a valid and frequently correct outcome.
11. **Reintegration.** Write accepted claims, evidence, contradictions, limitations, and questions back into the governed stores with the approval record attached.
12. **State comparison.** Document how C_(t+1) differs from C_t and how future inquiry should consequently change.

Steps 1 and 12 are the load-bearing pair. A cycle executed without both is ordinary AI-assisted writing.

7. The Structured Epistemic Change Record

A composite Reinforcement Learning Contour Differentiation Score would create false precision unless its components, coders, scales, and reliability were empirically validated. Worse, a composite score is a reward-hacking surface [23]: any dimension rewarding novelty will, under optimization

pressure, be satisfied by extreme claims. The framework therefore uses a structured categorical record rather than a single numerical index. Each dimension is documented separately. No total score is calculated, and no dimension may be optimized as a productivity target.

Dimension	Governance question	Permitted classification
Claim change	Which central claim changed?	none / refined / rejected / introduced
Evidence change	Did the quality or relevance of evidence change?	lower / unchanged / higher / contested
Contradiction	Was a serious counterargument discovered?	none / unresolved / partially resolved / resolved
Ontology	Did concepts or relations change?	none / local / structural
Uncertainty	Was confidence recalibrated?	increased / unchanged / reduced with justification
Inquiry policy	Will future questions or methods be selected differently?	no / possible / demonstrable
Provenance	Can the change be traced to evidence and decisions?	absent / partial / complete
External review	Was the claimed change independently examined?	no / internal only / external

7.1. Note on the Uncertainty Dimension

Justification is required for *reduced* confidence, not for increased confidence. An increase in uncertainty following contact with disconfirming evidence is ordinarily correct calibration and requires no defence; a reduction in uncertainty is itself an epistemic claim and must be supported. A governance instrument that treats rising uncertainty as a defect will systematically reward premature closure, which is precisely the failure the framework is designed to detect.

7.2. Conditions Under Which the Record Is Invalid

The record is categorical, programme-relative, and rater-dependent. Its classifications are not comparable across research programmes, across raters without an agreement study, or across time if the anchors change. It documents a decision; it does not validate it. A record showing *Provenance: absent* or *External review: no* is a statement about the limits of what has been established, not evidence of change.

The record must not be aggregated into a productivity score, used to rank researchers, or presented as a validated quality metric for a manuscript. In evaluation settings, at least two independent raters should assess central dimensions and report inter-rater agreement. Most importantly, the record must not become an optimization objective: once a measure becomes a target, novelty and contradiction handling become reward-hacking surfaces [23].

8. Novelty and Anti-Recursion Controls

8.1. Textual Versus Epistemic Novelty

A new title, section order, or vocabulary does not establish scientific novelty. The system must compare claims, mechanisms, ontology, evidence, and conclusions with the prior corpus.

8.2. Claim-Level Novelty Classification

Each central claim should be classified as previously established, reformulated, extended, contradicted, integrated across domains, newly formalized, or genuinely new. A manuscript in which no claim exceeds *reformulated* should not be submitted.

8.3. *Prior-Art Search*

Before a construct is claimed as new, it must be searched for under other names and in adjacent literatures. This control is the one most often skipped and most often fatal at review. Section 2 of this paper is an application of it.

8.4. *Counterfactual Novelty Test*

The system asks: if this work had not been produced, which future question, decision, concept, method, or evidence relationship would remain unavailable? If the answer is none, RLC value is low irrespective of prose quality.

8.5. *Mechanism-Change Test*

The strongest criterion asks whether the output modifies the mechanism that will generate future research rather than merely adding another answer. Content that does not alter question selection, evidence weighting, or ontology represents weak differentiation by definition.

9. Failure Modes and Controls

9.1. *Source Hierarchy*

Admissibility runs in descending order: primary peer-reviewed evidence; official documentation and authoritative datasets; systematic reviews and meta-analyses; scholarly preprints; institutional reports; reputable secondary analysis; search summaries; unverified generated content. The last two categories are never terminal evidence and may function only as discovery objects.

Failure mode	Risk	Control
Automated paper inflation	The workflow optimizes document count rather than epistemic value.	Count governed research cycles, including justified non-publication, rather than papers alone.
Conceptual echo chamber	The system repeatedly confirms the author’s prior framework; recursive self-consumption degrades the representation [21].	Require adversarial retrieval and at least one serious competing explanation per cycle.
False novelty	Existing ideas are renamed as original constructs.	Perform terminology search, prior-art comparison, and claim-level classification.
Citation laundering	An AI summary is treated as if it were primary evidence; fabricated references enter the corpus [20].	Retrieve and verify the original source; preserve claim-to-source provenance; enforce the source hierarchy.
Graph corruption	Incorrect entity resolution creates false relations.	Use schemas, confidence thresholds, provenance, and

		human review of high-impact edges.
Temporal inconsistency	New findings contradict old relations without superseding them.	Store validity intervals and explicit qualification or supersession links.
Ontology proliferation	New labels accumulate without consolidation.	Apply synonym mapping, concept-merger review, and ontology governance.
Reward hacking	Novelty incentives encourage extreme or weakly supported claims [23].	Use no composite score (Section 7); balance novelty with evidence, calibration, falsifiability, and relevance.
Automation bias	Technical sophistication is mistaken for scientific reliability.	Expose uncertainty, failed checks, rejected hypotheses, and unresolved disagreement.
Illusion of understanding	Breadth substitutes for comprehension; the question space narrows unnoticed [22].	Track question diversity across cycles as a monitored outcome.
Loss of intellectual authorship	Human and machine contributions become indistinguishable.	Record contribution provenance at claim, decision, and paragraph level [8].
Reflexive self-validation	The system produces, detects, and evaluates the change it calls progress.	Separate generation from evaluation; use preregistered rubrics, blinded comparison, and external audits (Section 10).

9.2. Citation Verification Standard

Before acceptance, verify that the source exists; that authors, title, and year are correct; that the DOI resolves where applicable; that the cited passage supports the claim as stated; that the work has not been retracted; and that the wording does not overstate the evidence. Given documented fabrication rates in generated bibliographies [20], this check is mandatory and cannot be delegated to the generating system.

9.3. Uncertainty Representation

The architecture distinguishes fact, supported inference, weak inference, hypothesis, speculation, and unresolved contradiction. Each is recorded as an explicit status on the claim rather than inferred from prose hedging.

10. The Reflexive Validation Problem

The RLC contains a structural circularity. The same human-machine environment may generate an epistemic change, define the categories through which that change is represented, detect the change, and judge whether it is beneficial. Human veto does not eliminate the problem because the human may also be the architect of the system, the author of the source corpus, the beneficiary of a novelty claim, and the final evaluator.

This creates risks of metric self-confirmation, endogenous novelty criteria, author-system collusion, and confirmation through ontology design. A system can appear to have learned because it has redefined progress in terms that favor its own outputs. Longino's argument [6] identifies why this is not merely a procedural inconvenience: objectivity is a property of critical interaction among differently situated inquirers, and a single-researcher recursive system lacks that structure by construction.

The problem should be managed, not solved, through separation of roles. Generation and evaluation should use different prompts, agents, or reviewers; central claims should be compared blindly with earlier work; source verification should be independent of drafting; rejected changes should remain visible; and periodic external ontology and provenance audits should be conducted. For empirical studies, evaluation rubrics should be preregistered before the research cycle begins.

One consequence is accepted here as binding: any evaluation reporting favourable differentiation without independent, blinded assessment should be discounted, including evaluations reported by the present author. Section 12 is written to that standard.

11. Falsifiable Hypotheses

H1. *RLC-supported workflows will produce greater claim-level novelty than non-recursive LLM drafting when assessed by blinded reviewers.*

H2. *RLC-supported manuscripts will contain fewer unsupported claims and fewer source-to-claim mismatches.*

H3. *Temporal provenance representations will improve detection of superseded, contradicted, or qualified claims compared with static document retrieval.*

H4. *Mandatory Structured Epistemic Change Records will reduce semantic and claim-level repetition across successive manuscripts.*

H5. *Human-governed RLC workflows will show better confidence calibration than fully automated research-generation workflows, at a measurable cost in throughput.*

H6. *The value assigned to a research cycle will correlate more strongly with future-question quality than with manuscript length or generation speed.*

H7. *Contradictory external evidence will produce greater beneficial contour differentiation than confirmatory evidence when governance controls are active, and greater harmful differentiation when they are absent.*

H7 is the framework's sharpest exposure. If contradiction produces no measurable differentiation advantage under governance, the central mechanism is not doing the work claimed for it.

12. Pilot Study Design

A 30- to 90-cycle pilot can compare three conditions: conventional manual research; AI-assisted drafting without recursive state and provenance; and the full RLC architecture. The study should use comparable research questions and blinded expert assessment.

Primary outcomes should include claim-level novelty, citation accuracy, unsupported-claim rate, contradiction-detection rate, useful ontology change, quality of subsequent research questions, reviewer-rated contribution, and traceability of epistemic change. Secondary outcomes may include researcher cognitive load, graph error rate, source diversity, question diversity across cycles, number of rejected hypotheses, and the proportion of cycles that correctly end without publication.

At least two independent raters, blinded to condition, should assess all novelty and contribution outcomes, with inter-rater agreement reported. Rubrics should be preregistered before the first cycle. Absent an agreement study, no scored comparison between conditions should be reported.

A high pre-publication rejection rate should not automatically count as failure. A governed system may create value by preventing weak, redundant, or unsupported manuscripts from entering the literature. This must be stated as a hypothesis-consistent outcome in advance rather than rationalized afterwards.

13. Discussion

The RLC changes the unit of analysis from the document to the transformation of the system that generated the document. This shift has practical and philosophical consequences. A research cycle may be successful even when it produces no publishable paper, provided it rejects a false assumption, exposes a contradiction, narrows an unjustified claim, reorganizes the ontology, or creates a more productive future question.

Prior publications accordingly function as evolving scientific memory rather than a static archive, including the rejected and falsified material that conventional archiving discards precisely when it is most useful for calibration. External search acquires an adversarial obligation: its function is to challenge the RLC, not to enlarge it, and a discovery pass returning only support has failed. Knowledge representation must be temporal, because belief revision is the phenomenon of interest and a store holding only current beliefs cannot represent it.

The framework also clarifies the distinction between automation of science and recursive scientific learning. Agentic research systems often optimize the execution of literature review, hypothesis generation, coding, experimentation, analysis, and drafting. The RLC instead asks whether the episode beneficially changes the epistemic conditions governing the next episode.

The system therefore should not be obligated to publish daily. It may complete one governed epistemic cycle per day. Possible outputs include a manuscript, a contradiction report, a negative result, an ontology revision, a literature map, a rejected hypothesis, or a reasoned decision not to write a paper.

Under this model, restraint is not absence of productivity. It may be evidence that the RLC has learned to distinguish document production from scientific contribution.

14. Limitations

1. The RLC is a conceptual construct that has not yet been empirically validated; the pilot described in Section 12 has not been conducted.
2. Beneficial differentiation is multidimensional, contestable, and often visible only over time. The Structured Epistemic Change Record documents decisions; it does not validate them.
3. The system's evaluator is not fully independent from its generator, creating the reflexive validation problem described in Section 10. This is the most serious limitation and is not eliminated by the controls proposed.
4. The use of the term *reinforcement learning* may still create technical expectations despite the explicit distinction drawn in Section 3.1.
5. The architectural components rely on probabilistic systems that cannot guarantee truth, novelty, or citation accuracy [19,20].
6. Temporal graphs remain vulnerable to extraction errors, entity-resolution failures, and misleading provenance.
7. A corpus centered on one researcher can preserve that researcher's biases even under adversarial search, because the search itself is framed from within the RLC.
8. Daily iteration may privilege rapid conceptual production over slow empirical validation, reintroducing through its own cadence the imbalance the framework opposes.

9. The proposed local implementation has not yet been compared experimentally with alternative workflows.
10. Extension to laboratories and institutions would require formal authorship, access-control, conflict-resolution, and governance mechanisms not specified here.

15. Conclusion

This paper introduced the Reinforcement Learning Contour as a governed epistemic feedback architecture for AI-assisted scientific inquiry. The term does not describe a conventional reinforcement-learning algorithm. It denotes the evolving configuration of knowledge, ontology, provenance, retrieval, evaluation, policy, memory, tools, and human authority through which research experience shapes future inquiry.

The framework is grounded in a simple proposition: the value of lived experience is proportional to the beneficial differentiation it produces between the preceding epistemic contour and the contours that govern future perception, interpretation, decision, and action. The Reinforcement Learning Contour names the governed architecture within which that differentiation is produced, recorded, and judged.

Applied to science, this means that the value of a research cycle is not determined primarily by how much text it produces or how quickly it produces it. Its value depends on whether it improves the mechanism from which future science will emerge.

A recursive research system becomes scientifically meaningful only when accepted outputs change not merely what the system says, but what it can subsequently perceive, question, test, reject, and understand.

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